

Manufacturing Practices

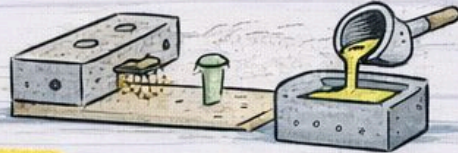
Module-1 Notes

by pyqfort.com



Contents Covered:

- Mould Making Intro

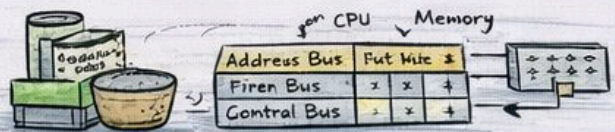


- Constituents of Moulding Sand

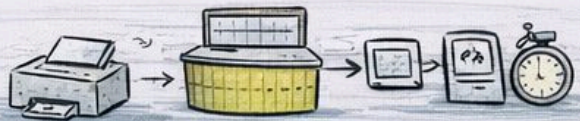


- Sand Preparation

- Properties and Types



- Testing of Moulding Sand

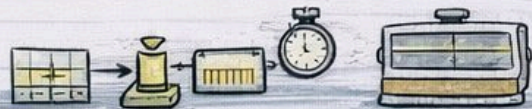


- Types of Moulds

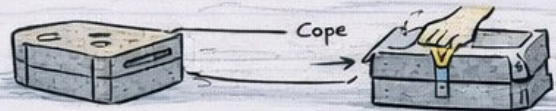
- Moulding Boxes



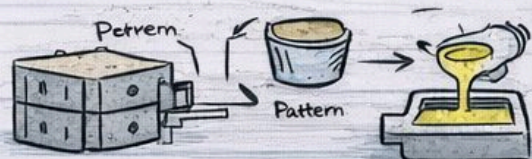
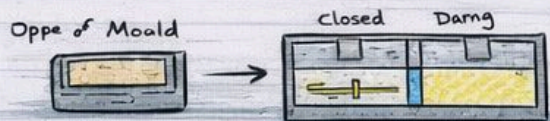
- Moulding Processes



- Moulding Boxes



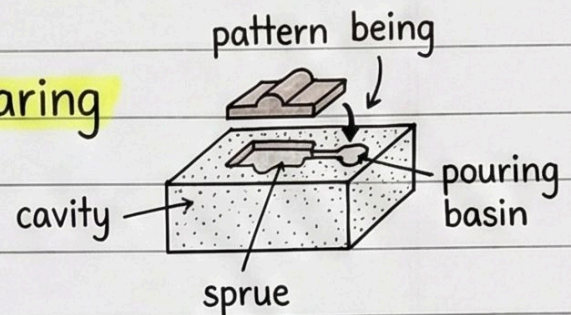
- Moulding Processes



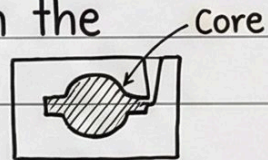
MOULD MAKING PRACTICE: INTRODUCTION & MOULDING SANDS

1. Introduction (Foundry):

- Foundry is the practice of preparing moulds for metal casting.



- Moulds are cavities made by the use of patterns.
- Moulds can be made of sand, plaster, or metals, but sand moulds are most popular.
- Additional elements in the mould: gating system, pouring basin, sprue, runner, etc.
- Cores are placed to get holes and recesses in the casting.



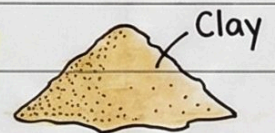
2. Moulding Sands:

- Common sources: rivers, lakes, sea, and deserts.

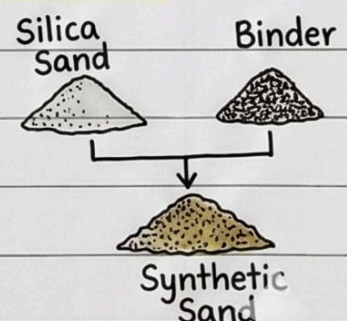


- Mainly grouped as:

(a) Natural Sand: Contains sufficient amount of binding clay. No more binder is required.



(b) Silica Sand: Does not possess the clay content. Needs addition of a suitable binder.

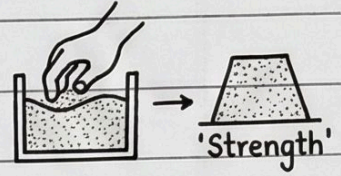


- When mixed with binders and additives, it is known as synthetic sands.

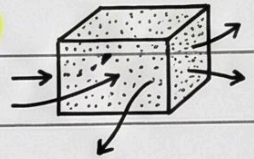
CONSTITUENTS OF MOULDING SAND: BINDERS

Purpose of Binders:

- Added to impart it sufficient strength and cohesiveness to the sand.



- Enables sand to retain its shape after mould is rammed and pattern withdrawn.
- Produces an adverse effect on the permeability of the sand mould.

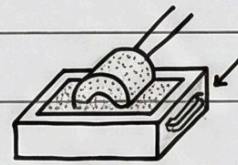


Classification of Binders:

- Grouped as: 1. Organic binders
2. Inorganic binders.

1. Organic Binders:

- Used specifically in coremaking.
- Common types: Dextrin, Molasses, Linseed oil, Cereal binders, Pitch (up to 2% max), Resins (phenol and urea formaldehydes).



2. Inorganic Binders:

- Common types: Clay, Sodium silicate, Portland cement.

- Clay binders are most commonly used.



- Types of clay: (i) Bentonite, (ii) Kaolinite, (iii) Limonite, (iv) Ball clay, (v) Fire clay, (vi) Fuller's earth.

- Bentonite is most widely used.

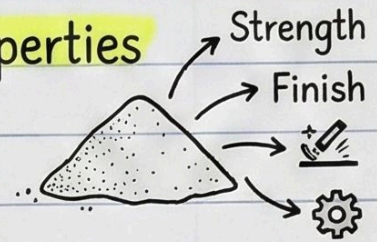


- Deposits found in Bihar, Rajasthan and Kashmir.

CONSTITUENTS OF MOULDING SAND: ADDITIVES

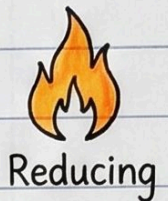
Purpose of Additives:

- Materials added to improve existing properties or impart new properties to the sand.



Commonly Used Additives:

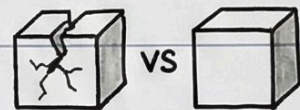
- Coal dust**: Used for grey & malleable iron. Creates reducing atmosphere, prevents oxidation. Reduces cohesiveness & strength.



- Seal coal**: Finely ground soft coal. Improves surface finish, restricts mould wall movement.



- Cereals or cornflour**: Promotes mould wall movement, reduces expansion defects. Improves collapsibility.



- Silica flour**: Increases hot strength, decreases metal penetration & expansion defects.



- Wood flour**: Promotes mould wall movement, improves collapsibility & thermal stability.



- Pitch**: Improves hot strength & surface finish for ferrous castings.



- Dextrin and molasses**: Increases dry strength.



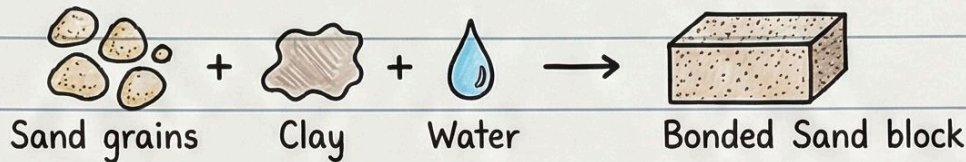
- Fuel oil**: Reduces the requirement of free water.



CONSTITUENTS OF MOULDING SAND: WATER

1. Purpose & Quantity:

- Water is added to activate the clay binder and develop the required strength and bond.



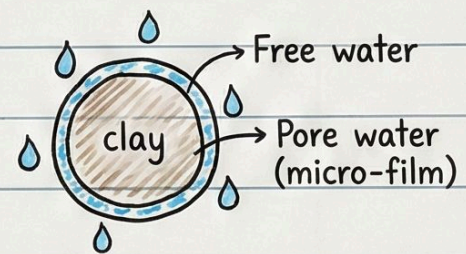
- Quantity varies from 2 to 8 percent depending on requirements.

2. Types of Water in Sand:

- Present in two forms:

(a) Free water: In the free state.

(b) Pore water: In mixed form, filling the pores of the clay.

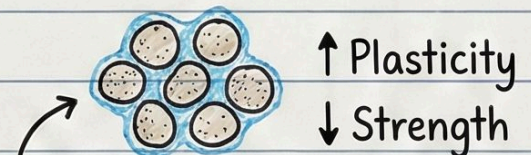


3. Role of Pore Water:

- Forms a rigid micro-film held by the clay.
- Mainly responsible for imparting strength to the sand.
- Bond quality depends on the thickness of this water film.

4. Role of Excess Free Water:

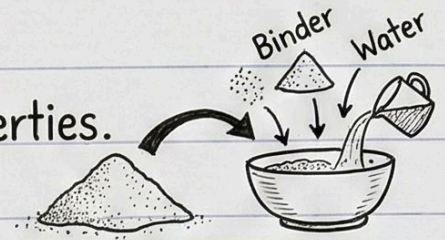
- Excess amount acts as a lubricant between clay particles.
- Improves mouldability and plasticity.
- However, it reduces the strength and weakens the mould.



SAND PREPARATION AND CONDITIONING

1. Why is it needed?

- Natural sand often lacks required properties.
- Sand mixing adds missing materials.
- Proper conditioning is crucial to avoid casting defects.



2. Goals of Conditioning:

- Uniform distribution of clay bond over sand grains.



- Even distribution and control of moisture content.

- Removing foreign materials (nails, gagers) by riddling.



- Thorough mixing of the entire mass.

3. Manual Conditioning (Small Foundries):

- Relies on moulder's practical experience (no testing equipment).

- Handful Test for judging condition:



- Grip and release a handful, break into two.

- No deformation at edges → Properly conditioned.



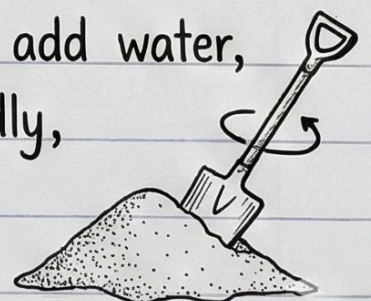
- Surface settling/compressed → High moisture.



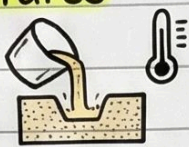
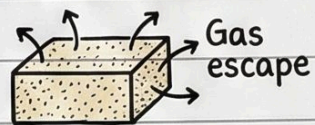
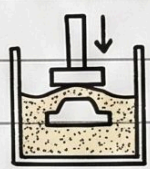
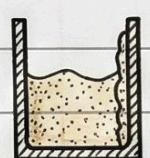
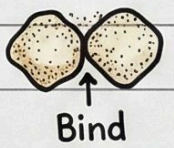
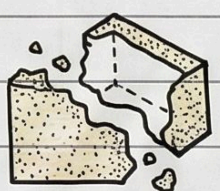
- Sprinkled grains/separation → Weak bond, low moisture.



- Mixing Process: pile sand & constituents, add water, turn upside down with a shovel repeatedly, then riddle.



PROPERTIES OF MOULDING SAND AND THEIR CONTROL

1. **Refractoriness**: Ability to withstand high temperatures of molten metal without fusing.
 - Depends on metal poured (e.g., steel > iron > brass), quartz content, grain shape & size (rough, larger grains → higher refractoriness).
 - Reduced by impurities (iron oxides, calcium, etc.).
2. **Permeability (Porosity)**: Ability to allow gases and steam to escape from the mould.
 - Crucial for sound castings.
 - Depends on grain size, shape, moisture, and clay content.
3. **Flowability (Plasticity)**: Ability to flow under ramming pressure to pack around the pattern.
 - Increases with clay and water addition.
4. **Adhesiveness**: Ability to adhere to surfaces of other materials (flask, gagers).
5. **Cohesiveness**: Ability of sand particles to bind together firmly.
 - Provides green strength and dry strength.
 - Depends on grain size, clay, and moisture.
6. **Collapsibility**: Ability to collapse automatically after solidification to allow metal contraction.
7. **Other Requirements**: Cheap & available, low coefficient of expansion, reusable, chemically inert, non-sticking to casting surface.

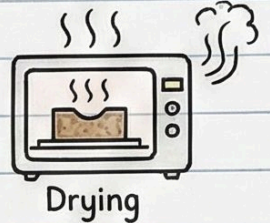
TYPES OF MOULDING SAND

1. Green Sand : Also known as **tempered sand**.

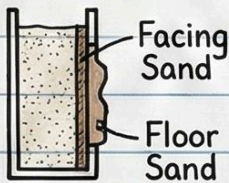


- Contains just **enough moisture** for **sufficient bond**.
- Used for **green sand moulds**, **no baking** required before pouring.

2. Dry Sand : Originally has **excess moisture**, which is then **evaporated** by **drying** the mould in a **suitable oven**.



3. Facing Sand : Also known as **'fat' sand**.

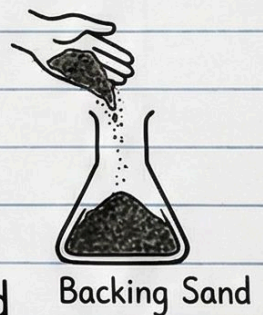
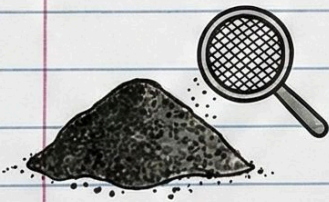


- Forms the **face of the mould**, rammed against the **pattern surface**. It is **fresh** and **well tempered**.

4. Parting Sand : **Sprinkled** on **parting surfaces** to **prevent sticking** of sand or pattern.
- **Burnt sand** and **dry silica sand** are used.

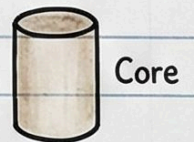


5. Floor, Black or Baking Sand : **Used sand** left on the **floor** after casting. **Riddled** to remove **foreign material**, then **re-used** to fill the **bulk** of the flask (**backing sand**).

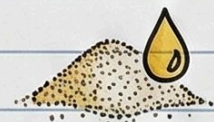


- Reconditioned floor sand in mechanised foundries is **unit sand**.

6. Core Sand : Has **high silica content**, used for making **cores**.



7. Oil Sand : Silica sand using **oil binders**.



8. Molasses Sand : Uses **molasses** as binder. For small castings with **intricate shapes** and **thin sections**.



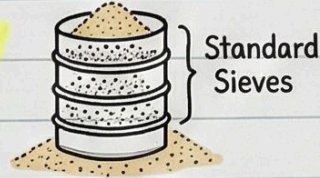
TESTING OF MOULDING SAND

Carried out to control properties.

Chemical tests for impurities, Mechanical tests for physical properties.

1. Grain Fineness Test: Indicates average grain size and percentage distribution.

Uses a set of standard sieves graded by mesh.



2. Moisture Content Test: 50g of moist sand heated in an oven at 105°C to 110°C to dry.



Formula: % Moisture = $\frac{\text{Weight of wet sand} - \text{Weight of dry sand}}{50} \times 100$.

Instruments used: Moisture teller, Moistmeter.

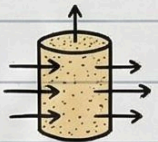
3. Clay-content Test: Sand is washed with caustic soda solution and distilled water to remove clay.



After settlement, clay-water mixture is siphoned off.

Process repeated until water is clear. Dried sand is weighed.

4. Permeability Test: Quantity of air that passes through a standard specimen under given pressure and time.



$$\text{Permeability No.} = \frac{v \times h}{p \times a \times t}$$

where v = vol of air, h = height,
 p = pressure, a = area, t = time.

5. Compression Strength Test: Finds maximum compressive strength of a standard rammed specimen (5 cm dia × 5 cm height).

Load is applied until breakage. Green strength is for moist sand.



6. Hardness Test: Uses a Mould-hardness tester (pocket instrument).



Principle is similar to Brinell, proportional to penetration depth.

TYPES OF MOULDS

1. Green-Sand Moulds: Made from moist sand, used for ferrous & non-ferrous alloys.

- Least expensive, time-consuming. No drying or baking.
- Not very strong, can be damaged during handling/pouring.
- Potential for blow holes/pinholes due to poor permeability.
- Cannot be stored for long.



2. Dry-Sand Moulds: Made from sands with binders, no moisture required.

- Surfaces sprayed with molasses, dried in ovens at 200°C – 300°C.
- Stronger, less damage. Moisture defects eliminated. Used for steel casting.
- Expensive, labour oriented, time consuming.



3. Skin-dry Moulds: Green sand moulds with moisture from surface (≥ 25 mm depth) dried using heaters.

- Combined advantage of green & dry sand moulds.
- Not as strong as dry sand moulds. Water can seep through dry layer.



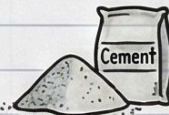
4. Loam Moulding: Constructed of porous bricks cemented with loam mortar.

- Loam mortar: Equal amount of sand grains & clay mixed with water.
- Brick structure faced with loam layer (6–12 mm) for shape.
- Used for large castings.
- Disadvantages: Requires large space/floor area, less accurate, time consuming. Not for repeated casting.



5. Cement-Bonded Moulds: Made by bonding silica sand with portland cement, dried in air. No backing.

- Advantages of dry-sand moulding, used for large ferrous castings.
- Mould materials cannot be reused, requires large space for air drying.



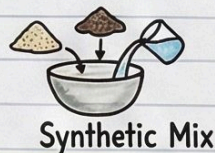
6. Oil-Sand Moulding: Sand, dextrin, bentonite mixture binded by linseed oil, vegetable oils, etc.

- Basic materials are very expensive.



7. Synthetic Sand Moulds: Mixed ingredients with sand and clay (Bentonite & water added).

- Advantages over natural sand: More uniform grain size, easier control of properties, less binder, high refractoriness, less moisture.

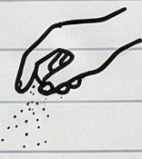
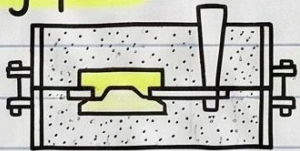
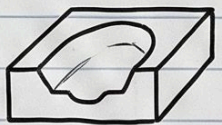
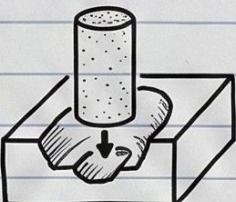
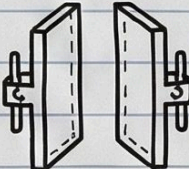


8. Metallic Moulds: Permanent moulds used for die casting.

- Metal introduced under air pressure.
- Presses operated pneumatically or hydraulically.
- Advantages: Less floor space, high production rate, quality casting, thin casting, good surface finish, precision.
- Disadvantages: High cost of moulds/equipment, special skills needed, limited range/size (non-ferrous alloys).



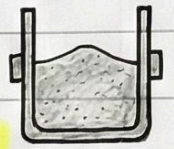
MOULD MAKING WITH THE USE OF A CORE

1. Select a **moulding box** of suitable size. 
2. Place the **drag part** upside down on the bench.
3. Position the lower **pattern** in the drag. 
4. Fill with prepared **green sand** and **ram** properly. 
5. Remove excess sand with a **strike-off bar**.
6. Sprinkle **parting sand**, then turn the drag upside down. 
7. Place the **cope** over the drag, align with **locking pins**.
8. Place the top **pattern**, **runner**, and **riser**. 
9. Sprinkle **parting sand**, fill the cope with green sand and ram.
10. Remove excess sand, form a **pouring basin**. 
11. Use **venting wire** to create escape passages for gases. 
12. Remove runner/riser, then separate cope from drag.
13. Remove the **patterns** to form the **mould cavity**. 
14. **Repair** any damage, blow off loose sand.
15. **Backwash** the cavity with **graphite/clay water** mixture for finish.
16. Bake the mould if required.
17. Place the **core** in the **core prints**. 
18. **Lock** the cope and drag together with **locking pins**. 
19. The mould is **ready for pouring**. 

MOULDING BOXES (FLASKS)

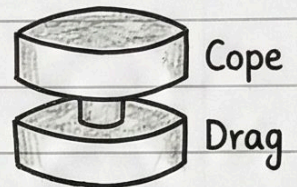
1. Purpose & Types:

- Moulding boxes or flasks contain the sand mould.
- Used in sand moulding, can be made of wood, cast iron or steel.
- Two main types: (a) Closed moulding boxes, (b) Open type of snap flasks.



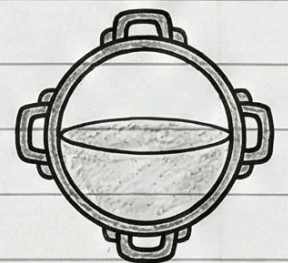
2. Parts of a Flask:

- Consists of two or more parts.
- Lower part is called drag.
- Upper part is called cope.
- Intermediate parts (if used) are called cheeks.
- Equipped with clamping means for pouring.



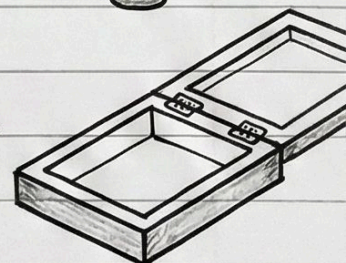
3. Materials & Use:

- Wooden flasks: Generally for green sand moulding.
- Metallic boxes (Cast iron/Steel): Required for dry sand moulds (heated for drying).
- Large, heavy boxes are metallic, handled by cranes.



4. Snap Flask:

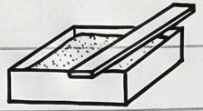
- Hinged in one corner, rectangular, made of wood.
- Used for bench moulding.
- Allows making multiple moulds from the same set of flask.



HAND TOOLS FOR MOULDING

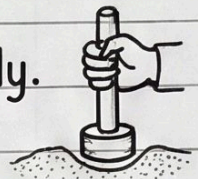
1. Sand Preparation & Handling:

- **Shovel**: For transferring sand from one place to another.
- **Hand Riddle**: To remove the lumps and foreign particles from the sand.
- **Strike off Bar**: To strike off the excess sand from the top of the moulding box.



2. Ramming Tools:

- **Rammer**: For packing sand in the moulding box uniformly.
- Two types: (i) **Pin rammer** (for narrow spaces) and (ii) **Flat rammer** (for flat surfaces).



3. Finishing & Repairing Tools:

- **Trowels**: Used for finishing flat surfaces and joints in the mould.
- **Slicks**: For repairing and finishing the mould surfaces and edges. Double ended.
- **Lifters**: For removing loose sand from inside the mould cavity.
- **Plane Smoother**: For smoothening the sand surface after striking off.



4. Venting & Withdrawal Tools:

- **Vent Wire**: Used to make small holes to allow the exit of gases.
- **Swab**: To apply a small amount of water around the pattern edge before removing.
- **Draw Spike or Screw**: To withdraw the pattern from the mould.
- **Bellow**: To blow out the loose or unwanted sand from the cavity.



5. Gating & Reinforcement Tools:

- **Sprue Pin**: Embedded to produce a runner for molten metal.
- **Gate Cutter**: To cut gates for easy flow of molten metal.
- **Gaggers**: Steel wire to reinforce the sand and support hanging portions.



MOULDING PROCESSES

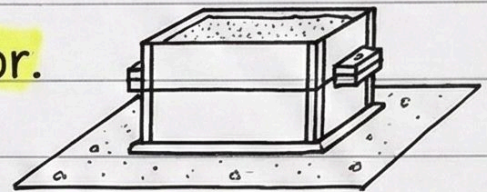
1. Bench moulding:

- For small moulds made on a convenient bench.
- Used in laboratories and workshops of polytechnics & colleges.
- Process is very slow and laborious.
- Uses hand ramming and loose patterns.



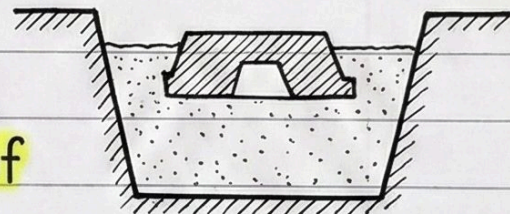
2. Floor moulding:

- Carried out on the foundry floor.
- For medium and large moulds.



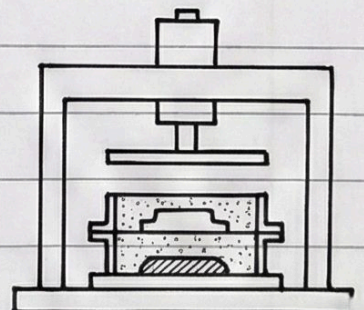
3. Pit moulding:

- For very large castings.
- Very big moulds are made in a pit or cavity cut in the floor of the foundry.



4. Machine moulding:

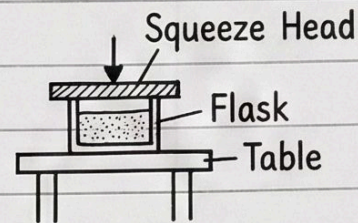
- Uses various types of machines.
- Makes small, medium and large moulds.
- Process is very fast and produces uniform moulds.
- Best suited for production work.
- Requires mounted patterns.



MOULDING MACHINES

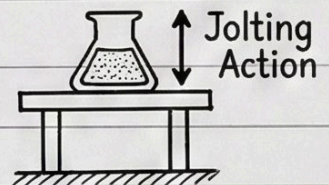
1. Squeeze Machine:

- Moulding sand is squeezed between the machine table and overhead squeeze head to attain desired density.
- Operated pneumatically or hydraulically.
- Two types:
 - Top Squeezer: Table is raised to compact sand against the stationary squeeze head.
 - Bottom Squeezer: Squeeze head is used to compact the sand. Density is not uniform.



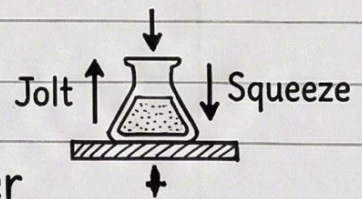
2. Jolt Machine:

- Flask is raised and dropped in succession (jolting). Sand is packed by sudden change in inertia.
- Sand packs evenly around the pattern, but is less dense at the top.



3. Jolt Squeeze Machine:

- A combination of jolt and squeeze actions for even ramming.
- Has both squeeze and jolt actions one after the other.



4. Sand Slinger:

- Consolidation and ramming obtained by the impact of sand falling with high velocity.
- Sand is thrown by an impeller at a rate of 500 kg/min to 2000 kg/min.
- Used for medium to large sized moulds; produces strong moulds.

